

deviation. As can be seen in equation (8.2), the z-score variable may be used to normalize the data that is otherwise unstructured.

$$c'_i = \frac{c_i - \bar{Y}}{\text{std}(Y)} \quad (8.2)$$

where c'_i represents mean normalized scores using the Z-scale c_i indicates value of the row Y of i^{th} column

$$\text{std}(Y) = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (c_i - \bar{Y})^2} \quad (8.3)$$

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n c_i \text{ or mean value} \quad (8.4)$$

Let's pretend we have a table with five rows, A , B , C , D , and E , and that each of these rows has a column labeled "I," denoting an independent variable. Thus, the z-score method may be used to find the standard data in each of the rows above (Rani et al., 2023).

Let's pretend that the standard deviation of a certain row is zero and that each value in that row is permanently set to zero. It is comparable to min-max normalization in that the z-score displays a range from 0 to 1. Decimal scaling is used, which allows for values between 1 and +1. This method assumes that

$$c_i = \frac{c}{10^Q} \quad (8.5)$$

Here

c_i illustrates the scaled values

C denotes the range of values

Q indicates the smallest integer $\text{Max}(|c'|) < 1$

To remove duplicates, we normalize the data as part of the preprocessing stage. The dataset tells whether the employee is departing or remaining based on the characteristics. To develop a prediction model, we now utilize 80% of the preprocessed data for training and 20% for testing. PCA is used to extract the key characteristics.

8.3.3 FEATURE EXTRACTION USING PRINCIPAL COMPONENT ANALYSIS (PCA)

Identifying all of the correlations between variables in data analysis might be tough. PCA transforms a great deal of information included in initial data sets into a series of important components, allowing for the discovery of hidden connections, enhanced information visualization, feature extraction, and categorization in new definition dimensions.

If learning algorithm is needed on the dataset, PCA can be quite useful, because it aids in effectively initializing coordinates for grouping. As feature extraction the PCA output, the first is initially normalize the dataset upon a unit scale (mean = 0 and variance = 1), which would be a condition again for maximum efficiency of several neural network models.

Our goal is to convert the p -dimensional database x together into different total sample y with a lower size (L_p), wherein x is just the principal component of

$$x = TD(y) \quad (8.6)$$

With y containing a collection containing n dimensions.

a. Organize into given data:

With y containing a collection containing n dimensions.

b. To use the equations, take the average:

$$\bar{y} = \frac{\sum_{j=1}^m y_j}{m} \quad (8.7)$$

c. Determine variability:

$$R^2 = \frac{\sum_{j=1}^m (y_j - \bar{y})^2}{(m-1)} \quad (8.8)$$

d. Determine the correlation coefficients:

$$Y^{m \times m} = (y_{i,j}, y_{i,j} = \text{cov}(\text{Dim}_j, \text{Dim}_i)) \quad (8.9)$$

e. Determine the principal components:

A principal component and natural frequencies of a matrix are at the heart of a PCA. Its principal components would define the major feature design's orientations, whereas the principal components would decide its intensity. When PCA is again a matrix, because if BY is an integer complex number of y , a non-negative variable x in n is called an equal zero of B .

$$BY = \lambda y \quad (8.10)$$

A number is referred to as a component of B , as well as the matrix y was referred to as a basis vectors equal with λy . Because the positive integer matrices that meet the equations are the principal components related to a component of matrices B .

$$(\lambda I - B)y = 0 \quad (8.11)$$

For the equivalent evaluation spaces, let model identify the collection B like all matrices y that meet equation (8.7).

8.3.4 PREDICTION USING THE EVOLUTIONARY GRADIENT BOOSTED RANDOM FOREST ALGORITHM (EGB-RFA)

The EGB-RFA typically uses a uniform voting weight across all decision trees when making a categorization. However, there is a fatal error in its application to classification predictions using unbalanced data. To address this concern, we have included the weighted F-measure into the EGB-RFA method.

By giving several decision trees varying amounts of weight, this method improves upon previous attempts at predicting employee turnover. From a data mining standpoint, the problem with predicting employee turnover is the need to classify the data into two categories, one of which is overwhelmingly negative.

We decided that “turnover” would be a good thing, while “not leaving” would be a bad thing. It’s not surprising that the negative group is larger than the positive one, which is in the secondary group. If the data is imbalanced, then measuring the model’s accuracy is meaningless. Although an accuracy rate of 96% is possible, it makes little sense in a circumstance where staff turnover is 2% and no one is expected to depart the organization.

Because we place such an emphasis on subcategories, the accuracy remains relatively high even if every subcategory is mistakenly labeled as a significant one. The model’s narrow emphasis on relatively unimportant factors makes it inappropriate for studying turnover in the workforce.

According to this research, it is preferable to incorrectly identify as a probable departure an employee who does not intend to leave the company than it is to fail to identify as a potential departure a person who does have the goal of leaving the company.

The effectiveness of each decision tree is evaluated using the harmonic mean of the F-measure, and the vote count is then calculated. Additionally, it integrates the two evaluation indices of accuracy and recall into a single measurement. The efficiency of the method for classifying imbalanced data is improved when compared to the EGB-RFA to updates used to the algorithm.

The F-measure is a combination of the outcomes of the accuracy and recall measurements, and the better the classification performance, the higher the F-measure. In the previous paragraph, in the part titled “Classification Evaluation Index,” we spoke of accuracy rate and the recall as shown in [Figure 8.3](#).

$$F - \text{measure} = \frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}} = \frac{2TP}{2TP + FP + FN} \quad (8.12)$$

Step 1: Verify that the training set, the validation set, and the test set are all correct. The bootstrap approach will be used to randomly extract $K+1$ datasets from the original dataset, each of which will have known categories. Each dataset has the same capacity of n , which is the same as the capacity of the original dataset. The first K sets from each of the $K+1$ datasets are put to use as training sets, and the set that is left over is put to use as the validation set.

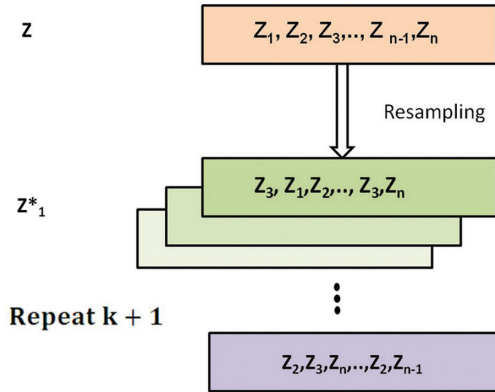


FIGURE 8.3 Schematic showing the bootstrap resampling technique.

The sample that has not yet been taken from data set serves as the test set and accounts for about one third of the entire sample that was included in the first training set. Figure 8.3 presents an outline of the bootstrap regarding the extraction approach with its accompanying specifics.

Step 2: Construct the radio frequency classifier. To construct the model, input K training sets and use the RF technique; next, use a combination classifier that is made up of K classification decision trees.

Step 3: Calculating the F-measure of the sub classifier will provide the weight value to use. Input the validation set, and then classify each sample in the validation set by thinking of each decision tree in the forest as an independent classifier. The precision rate, recall rate, total number of correct classifications, total number of incorrect classifications, total number of false positives, and total number of false negatives are all obtained. Next, compute the F-measure, which will equate to the weight of each sub-classifier.

Step 4: The performance of the model may be evaluated based on the input of the test set.

Step 5: Please feed in the samples that have not yet been categorized. Use the F-measure evolutionary random forest to determine how to categorize the data. The outcome, denoted by the character L , is determined by the percentage of votes received for each sub-classifier's classification findings.

The conclusion of the classification process might be stated as (8.14).

$$Y_i = E_i = \frac{2TP_i}{2TP_i + FP_i + FN_i} \quad (8.13)$$

$$L(z) = \arg_w \max \sum_{i=1}^k Y_i J(l_i(z) = W) \quad (8.14)$$

The combined classification model that was created by the EGB-RFA is denoted by the symbol $L(z)$ in equation (8.14). W is representative of the output variables (i.e., the classification type), l is the sub-classifier (also known as a single decision tree), and function is the indicator function.

In the question about the forecast of employee turnover, the parameter w might take either the value “leaving” or “not leaving.” Therefore, in (8.14), where w stands for departing, the sum of all the weighted values of the sub classifier that are categorized as turnover will be used as the score for $L(z)$.

On the other hand, when w indicates that an employee is staying put, the score for L will be calculated by adding together all of the weighted values of the sub-classifier that have been categorized as staying put (Z).

By comparing the two scores and the W value that corresponds to the maximum values of the two scores, the combined classification model predicts classification values. A representation of the structure of the EGB-RFA is shown in Figure 8.4.

The initial dataset, which contains historical information, is randomly segmented into a training set and a validation set. In addition, the data that were not picked are utilized for the test set. In the first step of the computation, the EGB-RFA algorithm is used to determine the relevance of each feature based on its position in the training set (Rana et al., 2021).

The factors that have been determined to be the most significant contributors to employee turnover are chosen. The second calculation involves incorporating these m characteristics into the EGB-RFA algorithm together with the training data in order to construct the prediction model for employee turnover.

Calculating the F-measure of each tree with the validation set in order to get its weight in the EGB-RFA method is how one obtains those weights. It is possible to forecast an employee’s desire to leave their position in the future by gathering many different characteristics about the individual and then feeding those data into a prediction model (Khanh & Khang, 2021).

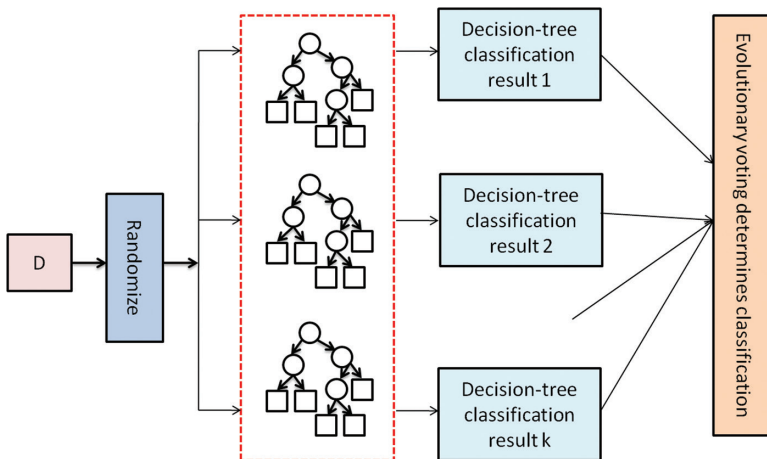


FIGURE 8.4 A representation of the structure of the EGB-RFA.

The accuracy, sensitivity, prediction error, recall, and F-measure of the prediction model may be determined by the use of a technique called 10-fold cross validation.

8.4 RESULTS AND DISCUSSION

We present a unique data-centric prediction model for the purpose of addressing these shortcomings by making use of the EGB-RFA. The dataset provided by a company’s HR department was used in the calculation of employee turnover (Khang et al., 2022a).

The performance metrics such as prediction error, recall, F-measure, accuracy, and sensitivity are analyzed. The proposed methods are compared with other algorithms: k-nearest neighbors (KNN), sequential selection algorithm (SBS), support vector machine (SVM), and decision tree algorithm (DTA) (Hajimahmud et al., 2022).

8.4.1 ACCURACY

The degree to which the result of a measurement complies with the appropriate value or a standard is what’s meant to be understood by the term “accuracy.” To get the greatest possible mark on a measurement, accuracy and precision are necessary. It is not necessary for a collection of measurements to be accurate in order for it to be precise, because a collection of measures may be organized according to their values as shown in Figure 8.5.

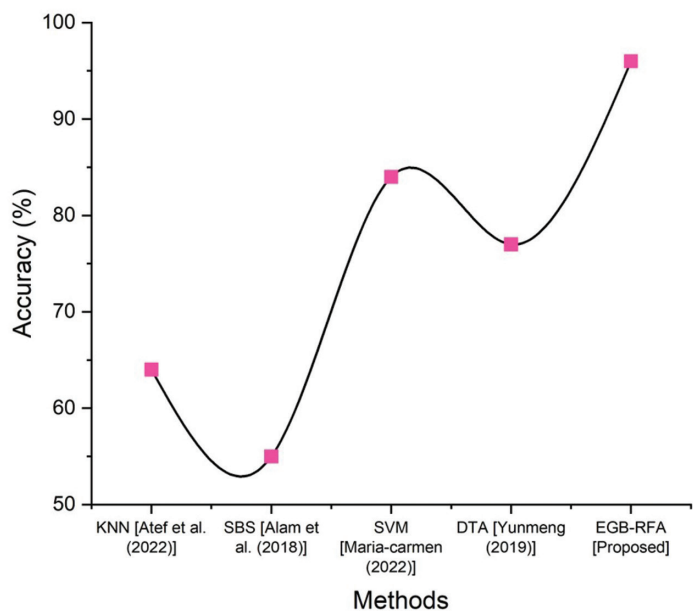


FIGURE 8.5 Accuracy for proposed and existing methods.

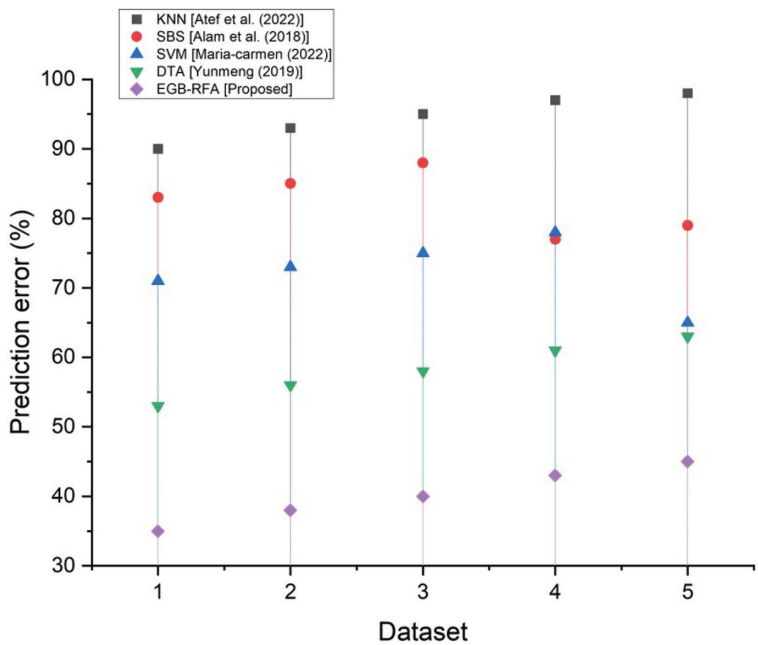


FIGURE 8.6 Prediction error for proposed and existing methods.

8.4.2 PREDICTION ERROR

The failure of a predicted event to occur is a prediction error. An error prediction plot compares the actual targets from the dataset to the expected values produced by the model. Consequently, we can determine the model’s variance as shown in [Figure 8.6](#).

8.4.3 RECALL

The proportion of instances that were correctly labeled as positive by the classifier is what is meant by “recall,” and it is used to establish completeness. The recall is a performance metric that is used in the process of selecting the best model, as shown in [Figure 8.7](#).

8.4.4 F-MEASURE

The F-measure is calculated as the harmonic mean of the accuracy and recall portions of the test. It is a statistical measurement that is used in the process of rating performance. To put it another way, an F-measure is a mean of an individual’s performance, and it is based on two factors: recall and accuracy, as shown in [Figure 8.8](#).

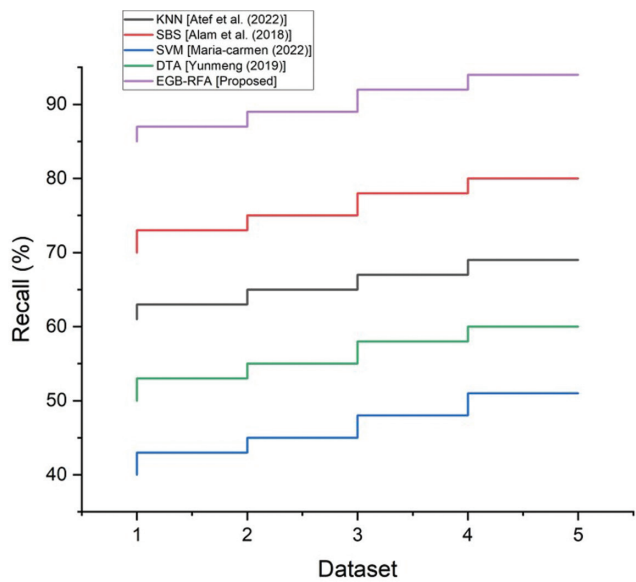


FIGURE 8.7 Recall for proposed and existing methods.

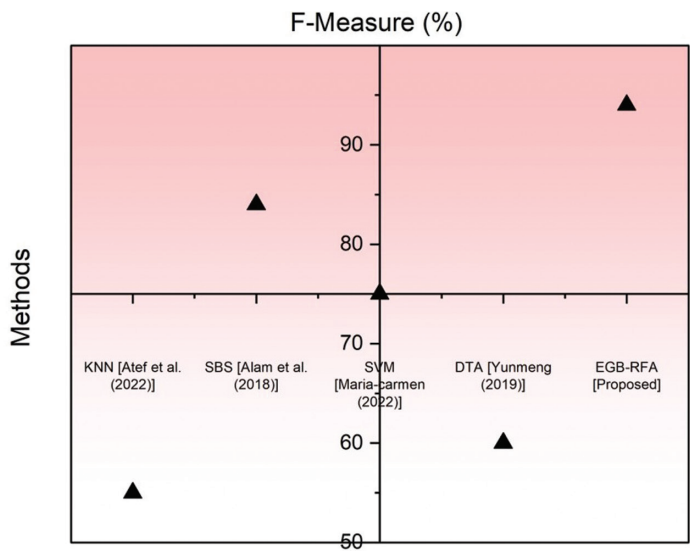


FIGURE 8.8 F-measure for proposed and existing methods.

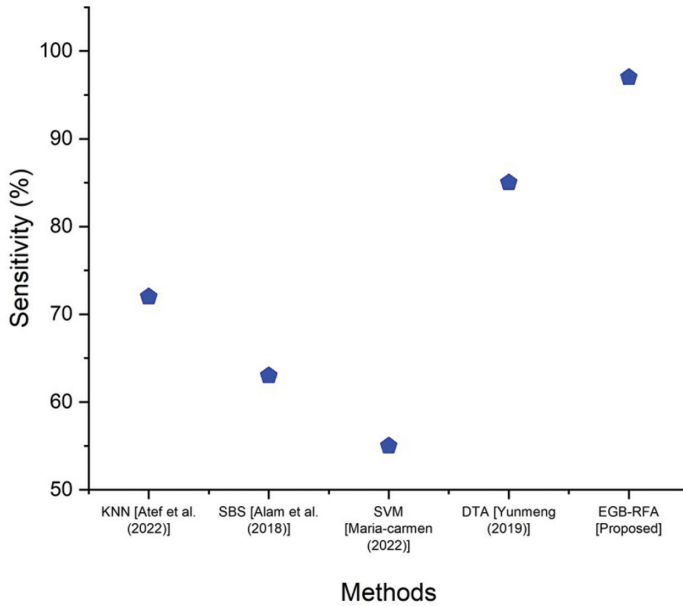


FIGURE 8.9 Sensitivity for proposed and existing methods.

8.4.5 SENSITIVITY

The degree to which a predicting model is able to recognize good examples is referred to as its sensitivity. In certain circles, it is referred to as the “true positive rate,” or recall. Because it enables us to observe how many positive cases the model was able to accurately detect, sensitivity is a factor that is considered when evaluating the performance of a model.

8.5 CONCLUSION

This work proposes the EGB-RFA, an enhanced RF algorithm based on the EGB-RFA F-measure. The key concept is to adhere to two phases.

- Prior to reducing dimensions and ranking feature relevance, the random forest technique is utilized.
- Second, the random forest technique is employed with the chosen features, and each decision tree’s weights are determined using the F-measure values.

The EGB-RF algorithm significantly outperforms the KNN, SBS, SVM, and DTA in terms of a number of performance metrics, particularly recall and F-measure, in the domain of employee turnover predictions (Morris et al., 2023).

Using the EGB-RF algorithm, we were able to determine the most important parameters impacting employee turnover in the employee dataset of a Chinese communications firm's branch: monthly income, overtime, age, distance from home, number of years at the business, and percentage wage rise.

Consequently, raising employee compensation is unquestionably one of the best strategies to keep staff. Different corporate workers will have some overlapping traits as well as some unique qualities, but this does not influence how this approach is used.

A company may create a strategy and take action to lessen the probability of employee turnover if it can foresee which individuals will leave, address the need to swiftly find replacements, and make other modifications to keep personnel in important roles.

HR managers may estimate employee turnover better and intervene sooner by using the EGB-RFA technique. This approach may also be used to address the issue of imbalanced data categorization in other disciplines, such as customer churn prediction, cancer screening, and abnormal testing (Tailor, Pareek & Khang, 2022).

These issues have similar traits, such a concentration on a minority group and the proverbial maxim that it is "better to mis-check than to mis-detect." In general, the EGB-RFA method has a greater recall rate for a minority class when compared to other algorithms. Future work will include conceptually analyzing these fields' data features in order to better validate and improve the EGB-RFA.

When compare the modeling to other methods in terms of study constraints, integrating the feature sorting and weight calculation operations does result in a larger time cost. Future studies might examine how to increase operational effectiveness and overall forecast accuracy. Additionally, this technique is not appropriate for sectors with large turnover rates, since it is built for uneven data. More research must be done on how to make the algorithm more general.

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9 Impact of Artificial Intelligence (AI) on Talent Management (TM)

A Futuristic Overview

*Arpita Nayak, Ipseeta Satpathy, B.C.M. Patnaik,
Sukanta Kumar Baral, and Alex Khang*

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9.1 INTRODUCTION

Dave Waters, pioneer in technology, has stated that “Artificial Intelligence is the future and the future is here.” Artificial intelligence (AI) is an area of computer science concerned with allowing computers to do cognitive activities that would normally necessitate human intellect. While AI is not a new concept, recent developments in processing power and digitalization have significantly boosted its applicability in corporate contexts (Rana et al., 2021).

According to the *Oxford English Dictionary*, AI is “the theory and development of computer systems capable of performing tasks ordinarily requiring human intellect, such as sensory perception, voice recognition, decision-making, and language

translation.” The root of AI came to light by an American computer scientist and inventor John McCarthy, the phrase was first used by the cognitive scientist in their proposal since 1955 at the Dartmouth Meeting 1956, the first conference on artificial intelligence (Khanh & Khang, 2021).

Non-human intelligence is described as being created with specific goals and objectives in mind. The question was if it was feasible to create a machine that could comprehend abstract ideas, solve issues, and develop exactly like a person. Theoretically, he asserted, “any aspect or other attribute of intelligence may be represented so flawlessly that a computer may be built to mimic it. In the next 10 years, according to recent findings from a sizable poll of machine learning academics, AI will beat humans in various tasks, from language translation to surgery”.

Additionally, according to researchers, there is a 50% possibility that AI will surpass human performance in all activities in 45 years. AI systems are influencing nearly every part of our life to increase profitability and improve our human skills. It’s difficult to imagine living without artificial intelligence, since it has permeated every aspect of our daily existence.

The essence of AI has been affecting our lives so much that it is difficult to imagine a future without it, especially in business. AI can gather and organize huge volumes of data to draw inferences and make projections that go beyond what a person would be humanly capable of understanding (Bhambri et al., 2022).

Furthermore, it improves organizational effectiveness while lowering the likelihood of a mistake, among other things; it rapidly detects random patterns, including frauds and spam, to alert businesses to suspect behavior. In the modern world, AI research is continually developing. According to technology journalist Alice Bonasio (2022), the growth of AI research over the previous five years has been 12.9% each year.

We are just a few steps into the Fourth Industrial Revolution. Throughout the history of technology, from steam and water energy through electricity, industrialization, and now computerization, the notion of what it means to be human has been called into question (Khang et al., 2023). The Fourth Industrial Revolution combines humans, robots, and smart devices to enhance production processes (Tailor, Pareek & Khang, 2022). AI devices are gradually replacing monotonous, repetitive, and laborious human jobs, including scheduling, documenting, and checking equipment. They also increase data collection and preliminary analysis.

The Fourth Industrial Revolution’s potential to raise income levels and enhance living standards throughout the world is one of its most important advantages. Better goods and services for individual consumption and production will result from the use of smarter technology in our workplaces and industries.

In the current scenario, organizations cannot afford to ignore their employee’s needs. Employees are looking for certain companies which play a significant role in their career development and a workplace where their talents are being given the top priority (Snehal, Babasaheb & Khang, 2023).

Before AI integration, human resources (HR) had an important job and was expected to provide employees with fast assistance within the allotted time. Managing talent has been highlighted as a major duty for the HR department, but the accumulation of paperwork and the demand for quick turnaround times has presented the HR department with its greatest problem (Hajimahmud et al., 2022).

Companies questioned by the SHRM Society 10 years ago stated that succession planning and equipping managers with the necessary skills would be their biggest problems in the future (Babasaheb, Sphurti & Khang, 2023a).

The HR of the next 10 years must be more concentrated on talent management (TM), productivity, and performance, and taking quicker action. To do so, thinking must undergo a seismic upheaval.

AI incorporation into TM processes will benefit businesses, since these deployments can analyze, forecast, and diagnose data to help HR practitioners make better decisions. According to a study published in the *International Research Journal of Engineering and Technology*, AI can assist HR teams in making better-informed and productive decisions (Bhardwaj, 2020).

Since the turn of the twentieth century, significant socioeconomic change has been sustained because of the growth of automation, manufacturing technologies, and digital communications. Our online interactions, communications, and professional lives are now pervaded by AI systems and machine learning.

With advancements in AI science increasingly, analyzing human-related factors to meet the hiring the quality workforce, the human resources sector is necessarily moving toward total digitalization model. It's time HR gave these developments a serious look, as stated in the study that businesses must efficiently manage their HR through hiring, training, and employee development.

According to a report, the rise of AI and machine learning has allowed HR professionals to focus more on TM and strategic roles rather than administrative ones, necessitating their extreme adaptability to the constantly changing world (Khang, 2024).

The advising role, such as mentoring and employee motivation, will be more favored by HR professionals in the future. According to Khang et al.'s (2023c) research, technology will only eliminate the monotonous jobs from the HR function; for HR professionals to survive, they must acquire new skills, improve their capabilities, and TM process and view technology as an advantage rather than a menace.

Ngor, Channing & Margarita (2023) stated that compared to 48% in the US, 81% of Chinese firms questioned utilizing AI for personnel information searches. The majority of Chinese businesses want to spend money this year on chatbots for employee self-service.

Chinese IT giants Baidu, Alibaba, and Tencent are making investments in AI both domestically and internationally. The promising aspect of AI has witnessed significant growth in the use by HR personnel (Babasaheb, Sphurti & Khang, 2023b), whether it's the targeted approach toward recruitment, performance management, onboarding process of employees, or help in enhancing the way of managing the performance of employees. Nowadays, AI has helped in tracking, and maybe most enticingly, quicker and more efficient way (Gonzalez et al., 2019). In truth, for forward-thinking companies, the future is already here and is continually developing.

Without a doubt, these advances in TM are providing an effective platform for it. The focus on TM in HR is no longer a "battle for talent." According to the talent war, Company A employs a top-tier employee from company B, and company B recruits a top-tier person from company A, a win-win scenario. Current personnel management has progressed beyond this, and the debate is now about how to attract

and retain existing individuals by providing them with interesting work experiences, all thanks to AI (Mahfoozi et al., 2018).

Humans are excellent at learning but terrible at unlearning. The psychological processes that cause us to be biased are frequently the same skills we utilize to cope in our daily lives. The world is just too complicated for us to constantly comprehend rationally and purposefully; if we did, we would be overwhelmed by an information explosion and unable to make simple judgments, such as ordering a cup of coffee (and besides, why should we trust the barista if we don't know him?). That is why it is easier to guarantee that our data and training sets are impartial than it is to modify Sam's or Sally's behavior, from which we cannot eliminate bias or extract a list of the factors that impact the behavior (Gupta et al., 2023).

AI has made TM bias-free in certain corners, as it is the machines that are running the analysis, and the bias ratio is relatively much lower in short, because AI puts an ethical working environment (Evseeva et al., 2021) in TM as shown in Figure 9.1.

Dexcom and Micron are achieving considerable benefits by using a unified AI-powered people management tool that adjusts to their strategic plan and provides new insights.

AI-powered technologies provide chief human resources officers (CHROs) and HR professionals with new insights, intelligence, and advice, supporting them in reducing their organizations' growing talent gaps (Khang et al., 2023).

By merging acquisition, job rotation, inclusion and diversity contingency workforces, training and development, and performance management on a single AI platform, HR leaders gain superior insights into addressing personnel shortages. It's also

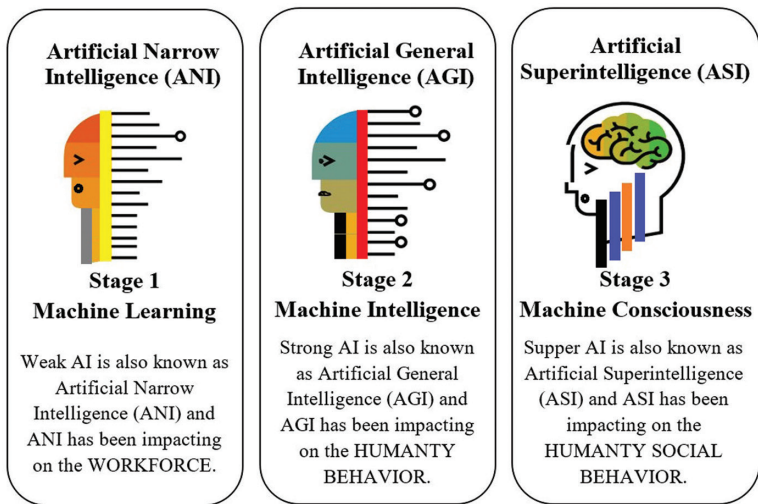


FIGURE 9.1 Illustration of different main stages and benefits of AI.

encouraging to see how AI technologies assess applicants based on their abilities while disguising flaws (Wiblen & Marler, 2021).

Organizations have been struggling to retain talent for more than a year. According to US Labor Statistics, 4.2 million people left their jobs voluntarily in August 2022. Simultaneously, there were 10.1 million job opportunities.

Conventional ways to attract bright individuals haven’t always worked in this intensely competitive market, as seen by the Great Resignation and more contemporary trends like “quiet quitting.”

9.2 IMPACT OF AI ON TALENT MANAGEMENT

An increasing wave of AI solutions for personnel management has the potential to assist firms in finding better prospective employees more quickly, providing more impactful staff growth, and promoting retention through more efficient employee satisfaction (Asplund, 2019), as shown in [Figure 9.2](#).

9.2.1 ARTIFICIAL INTELLIGENCE (AI) AND LEARNING AND DEVELOPMENT (L&D)

Learning and development (L&D) is a continuous activity that promotes the professional improvement of firm employees. It comprises identifying skill gaps in your job and developing training programs to offer employees the particular knowledge

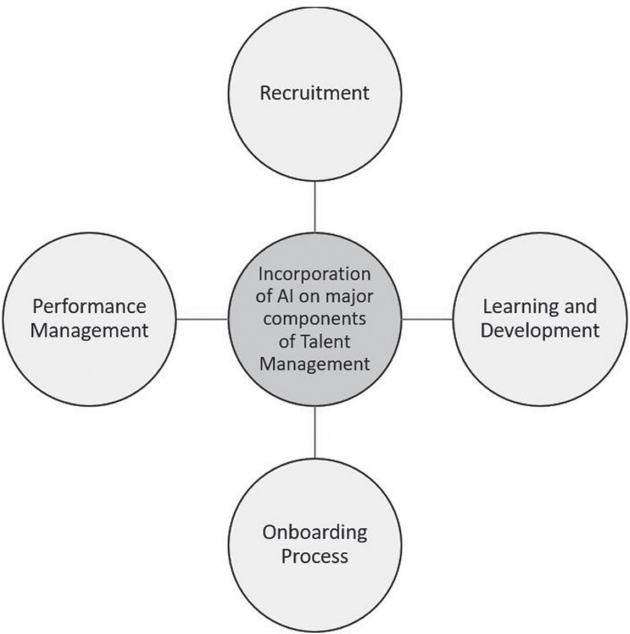


FIGURE 9.2 Study of the impact of AI on talent management.

and abilities required to increase production. L&D frequently focuses on reskilling or upskilling employees so they can fill new positions within the firm or execute their tasks more effectively. This helps firms recruit, develop, employ, and retain people.

Furthermore, it enhances the working atmosphere for employees, reduces turnover, and increases retention. The effect of AI in learning and development is more persistent, such as understanding the learning gaps (AI can calculate large amounts of data, integrate them, and pinpoint knowledge gaps in a person).

The creation of learner profiles, which can improve the overall learning experience and impart new skills, can then be done using these contradictory pieces of information. Employers may find it simpler and more affordable to train their staff in certain areas when upskilling them if they can pinpoint where they have knowledge gaps, and personalized learning styles (employees may organize their individualized learning materials, choose their objectives, and gather information regarding the learning preferences and styles) with AI in L&D.

They will be able to do this and achieve the standards established by the organization while learning the customized information at their speed. The most pivotal factor is time convenience i.e., assistance at all times. (It's typical for your employees to ask questions when learning online at their speed. During their employee training, your employees may obtain answers right away by using chatbot-supported AI technology. Because of this, answering the queries quickly allows your employee to continue with the L&D.)

According to Bank of America, artificial intelligence will be the driver of an annual industrial increase of \$14–33 trillion by 2025. To keep pace with changing needs, L&D will need to have access to a huge database, as it will help them to frame the L&D structure better (Wang & Wang, 2022). Use of AI has made it easier for workers to learn about many elements of their interests, since it saves time and provides prompt answers to questions while also enhancing their understanding.

Stephen Walsh, a cofounder of Andres Pink, wants to include learning in the daily workflow, while 56% of learning is quite formal and provided in person. The answer to this issue is a learning system that uses AI, and the advantages are the same: Time is saved, workers are active and engaged, the process of learning is automated, and the organization's profit increases at a super-rapid rate.

Completion rates will increase when training information is delivered by artificial intelligence in the desired learning format for the learner and is then reinforced by exciting techniques (Babasaheb et al., 2023a).

In this respect, several multinational corporations (MNCs), including Amazon, Infosys, IBM, and WalMart, are producing goods and services that are driven by AI while simultaneously investing in the development of the necessary L&D for employee skill enhancement (Khang, 2024).

These specialized algorithms allow your learning platform to do some of the operations that you would undertake manually as a learner or administrator. For example, your learning platform can now evaluate practically all of the different forms of information that you're contributing to it safely and securely.

Not only that, but it may also improve and streamline the learning process. Learners may utilize the platform's global search to identify the most relevant material for them, eliminating the requirement for an administrator to manually classify content.

In addition, based on their content history, learners can invite other learners to read their published content. The best aspect is that it all takes place behind the scenes, never interfering with the learning experience. According to Firm Mercer's Global Talent Trends report in 2019, 60% of businesses, including 59% in the US and 55% in China, want to increase the use of industrial automation this year (Ngor, Channing & Margarita, 2023).

In a survey conducted in Boston Consulting Group (BCG) and Munich Institute for IT Service Management (MITSM), a poll of 3,000 company executives produced some intriguing findings which showed that 75% of executives believe AI will enable their companies to access new markets, giving a significant opportunity to their employees to enhance their skills. Almost 85% of respondents believe AI will help their companies increase employee productivity and maintain a competitive advantage and learning development.

According to a blog titled "Impact of AI on Learning and Development (2022)", organizations have already begun using AI in their business and daily working process, and it is now the time for L&D personnel to adapt. Many organizations have already adapted AI in their L&D systems, which is advantageous to them in the manner of providing employees with personalized learning experiences, AI-based digital coaches, and an integrated way of training the employees, and as a result, it provides privacy and security (Khang et al., 2022).

To accept the existence of its new AI "employees," corporate culture must develop. Bots and humans will collaborate to educate and learn from one another. This type of collaboration, like successful human coworker relationships, necessitates trust and transparency. For example, when a division, function, or team performs badly, human labor may be inclined to blame the computers for poor performance or erroneous output rather than discovering and resolving defects, regardless of whether they are intentional or unintentional.

Human or computer and adversarial relationships, like human relations, will impede addressing common challenges and enhancing shared rewards. Employees need to be motivated to adopt a new mindset and try to help computers help those (Plastino & Purdy, 2018). According to Plastino and Purdy's (2018) survey, 92.6% of the 27 HR/training professionals asked a toy their organization requires an expertise management strategy, while 40.7% think training content delivered to a certain group of workers must be refreshed regularly. Sixty-three percent of respondents highlighted personalized learning as a must.

Overall, 92.6% of HR/training experts felt that training programs should promote a high level of employee participation. Overall, 51.9% of those surveyed preferred on-the-go tools for their employees, while 33.33% said an easy e-learning and development interface would be beneficial. (Tambe, Cappelli & Yakubovich, 2019)

9.2.2 AI's INFLUENCE ON RECRUITMENT

Steve Jobs, chief executive officer (CEO) and co-founder of Apple stated, "The secret of my success is that we have gone to exceptional lengths to hire the best people in the world." A 2017 Deloitte poll found that 83% of CEOs rated recruitment as vital or essential (Deloitte, 2017).

Nevertheless, due to the talent and skill gap, TM becomes just as crucial as other resources for approaches. Companies are keen to take essential actions for managing such talents; TM is a hot issue right now. Organizations are utilizing innovative hiring approaches to quickly fill talent shortages with a particular skill set. Organizations are using tactics to bring in a substantial amount of talent, bolstering the recruitment component (Deloitte, 2017; Whysall, Owtram & Brittain, 2019).

The AI sector had a startling \$1.2 trillion market value in 2018. Sixty-one percent of companies were allegedly employing AI in some capacity. Even small and medium-sized businesses, which account for 90% of all businesses and 60% of the workforce in the UK, are beginning to employ AI in their recruitment thanks to companies testing out AI products that are specifically suited to their niche market.

The Institute of Chartered Accountants in England and Wales (ICAEW, 2014) has stated in their study that AI technology is being used by HR managers for a variety of HR functions, from employee onboarding to workforce planning.

Organizations frequently utilize AI technologies for hiring employees. In talent intelligence systems, AI algorithms connect succession planning to your hiring procedure, enabling you to examine both internal and external candidates at the same time. Now that you have the right knowledge, you can confidently select the best candidate (Pillai & Sivathanu, 2020).

AI has grown significantly in relevance for managing talent, and it now exists at three levels: assisted intelligence, augmented intelligence, and autonomous intellect. It has helped in the task of recruitment in a structured way. AI technology enables collaboration between humans and machines to produce work decisions.

For instance, conversational AI-powered bots assist in developing customized conversational tone, providing applicants with realistic and engaging experiences across many channels, setting up interviews, responding to inquiries from candidates, and encouraging recommendations. The workplace is undergoing a complete revolution thanks to autonomous intelligence.

AI technology takes independent action and produces the outcomes. Subconsciously, it gathers and evaluates the data. For instance, AI offers candidate selection outcomes based on specific criteria.

AI technology is mostly used in recruiting, training, job involvement, and retention, which reduces costs and saves time while effectively completing HR tasks. By 2030, according to the McKinsey Global Institute model (McKinsey, 2017), almost 70% of businesses will have implemented AI in some capacity.

AI may identify patterns that are undetected by the human eye, in contrast to conventional recruiting techniques like curriculum vitae (CV) screening and in-person interviews. It might be utilized to locate the ideal candidate for the position more quickly and successfully than ever.

The hiring process is one of many CEOs' top concerns, according to various polls. The Society for Human Resource Management estimates that companies pay an estimated \$4,129 for each position in US hiring. A large-scale study showed that the establishments that have adopted AI in the system have benefited from AI in such contexts as big data analytics, predictive modeling, and automated evaluations.

With the effective use of AI, TM is converted from manual, repetitive processes to conventional ways with the help of technology, generating more efficient results

and saving the company's time and money. The quality of hires also improves, and the HR process is subsequently reformed by AI.

Even when you search for "artificial intelligence in recruiting" on the internet, you'll see phrases like "faster time to interview," "start making recruitment easier," and "Goodbye, traditional recruitment methods." These advantages are undeniably appealing to businesses.

Globalization, low unemployment, mechanization, significant technical advances in machinery, and other factors are all driving forces that have increased competitiveness for highly competent job candidates, but organizations that adopt AI may be concerned about lagging behind their competitors.

Human resources are a country's most precious asset and its first pillar of success. HR is an important part of a company for innovation, growth, and strategic edge, so organizations fight for highly skilled and competent human resources. HR receives a great deal of consideration in selecting methods to ensure its efficacy because of the significant cost of mistake recruitment (Failing to consider recruiting from within. Relying too much on the interview. Using unconscious bias. Hiring people less qualified than you. Rejecting an overqualified candidate) or the ability to assign an appropriate candidate for the position.

The motive of the selection and recruitment process is to offer the company plus suitable human capital at the lowest cost, adding attention to the essential activities and behavioral qualities required for work quality criteria, while still producing a comparatively flawless product. Making a selection is challenging.

The data-based search engine (Strohmeier & Piazza, 2015) is a system formulated to search for information on the internet and is one of the most frequently used AI tactics in recruitment. Search engines recognize the significance of the search term and conduct a web-based investigation for candidates who fit the job posting and profile.

According to Black and van Esch (2020) and Mochol, Jentzsch and Wache (2007), employers must specify keywords or a "normative framework" that explains various employment characteristics such as job title, credentials, competence, and so on.

To enhance search results for matched candidates, the knowledge-based search engine uses an ontology-driven information extraction within the system. (Van Esch & Black, 2019). The standard recruiting procedure, which includes a CV and interviews, has been determined to be unproductive. To mitigate this danger, numerous companies have established pre-employment examinations that ask applicants a set of questions. These are consistent (even when repeated) and genuine.

Technology has continuously assisted recruiters in screening more prospects and providing a higher-quality, more cost-effective service to both job seekers and employers. AI is expected to follow suit.

The recognition that AI is "game-changing for HR" shows that it might have a significant impact on the process of recruitment. According to a poll done by the HRP (2016), 84% of HR specialists at businesses said AI was a valuable tool in recruiting. AI facilitates real-time social activities, including bargaining, persuasion, and caring.

According to 85% of respondents in a recent poll on AI in recruiting, performance assessments, interview scheduling, finding candidates, and finally onboarding new employees would all benefit from AI (Hunkenschroer & Luetge, 2022; Kaplan & Haenlein, 2019).

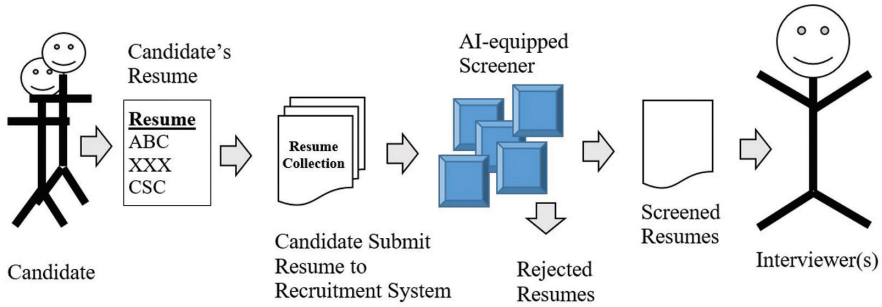


FIGURE 9.3 Illustration of AI in screening of curriculum vitae.

Individuals can also utilize performance assessments to determine which abilities they need to hone or gain to increase their job efficiency. Using advanced data, it is possible to tailor an employee's experiential learning. Talent seems to have a shorter shelf life than in the past.

AI can help the system determine which individuals need to reimagine their talents when they become obsolete or supplanted by newer technologies as shown in Figure 9.3.

9.2.3 THE IMPLICATIONS OF AI ON PERFORMANCE MANAGEMENT

Employees look for performance reviews to motivate them and help them improve, much like plants require water to thrive. Performance management is a continuous, data-driven conversation rather than a one-time discussion each the year, as Adobe demonstrated years ago.

A growing number of businesses are realizing the value of ongoing, in-the-moment feedback to guarantee that staff members get the constructive criticism they require as well as the positive feedback they deserve.

To comprehend future leaders and their growth potential, organizations must identify their abilities inside the organization. HR specialists studied that when it relates to recognizing high-performing employees and potential leaders, using raw data and insights created by algorithms, AI-powered systems like these may give a more impartial approach to performance monitoring, talent development, and employee appreciation (Buck & Morrow, 2018; Garg, Srivastav & Gupta, 2018).

A survey report by profit.co (Para 2) revealed that organizations started using AI mostly in the COVID-19 period, when our daily lives were suddenly halted. AI helped in providing an optimum, data-driven solution, as data gathering is constant and automated in an artificial intelligence-powered performance management method. As a result, performance evaluations might be continuing.

In addition to the material that has been obtained, AI analyzes a central library of information from past assessments and a variety of sources. As a result, erroneous judgments are removed, and employee performance evaluations are given more weight to help in the effective functioning of the TM team.

Based on Mehrotra (2021), the implementation of AI in performance management (PM) will result in real-time feedback, eliminating annual response instances and replacing them with more frequent data-driven conversations. This will make managing employee performance more future-oriented by emphasizing individual development rather than previous performance measurements. PM with AI is entirely data-driven. It doesn't have any preferences of its own. It gathers data from several sources, removing the possibility of data errors. Therefore, AI may assist managers in giving employees and the business unbiased input.

AI-powered performance evaluations sift through enormous quantities of data to provide forecasts that take into consideration the employee's career development and performance history in addition to the review's findings (Canals, 2020). This is essential for assessing an employee's ability and determining the most suitable goals. The manager may be better able to recognize advancements and determine rewards with its aid.

An AI-based system provides managers with immediate feedback and real-time insights into the performance of specific employees. Additionally, it saves a significant amount of time spent on performance review preparation. With AI, you can continually collect data from a variety of sources, such as calendars, employees' written correspondence, and other sources. All lead to effective management of talent.

The implementation and adoption of AI have helped in an efficacious way of managing talent. The major benefit of using AI in performance management is real-time insights (an AI-based system provides managers with immediate feedback and real-time insights into the performance of specific employees), and effective use of time (additionally, it saves a significant amount of time spent on performance review preparation), and data collection.

The implementation of AI has an effect on performance management in a way that increases its dependability through the use of more accurate and up-to-date data, facilitating the acquisition of performance information. The procedure becomes impartial (Alekseeva & Gine et al., 2020). A performance management revolution is now taking place. Companies are increasingly moving away from the typical yearly review, which is conducted once or twice a year, to more regular and casual check-ins twice a year.

Companies are becoming more aware of the requirement for employees receive feedback in real time. This brand-new approach no longer employs 360-degree feedback, which involves asking an employee's coworkers for input on a project employee's performance, since 360-degree feedback is not available, is considered the most impartial feedback tool available, and takes a long time.

However, many organizations adopt the use of performance reviews as a component of performance management. It is critical to discover a solution to shorten the management process required for the implementation and management of these reviews.

Companies recognized that they required something more agile, real-time, more personalized, and entirely focused on fueling performance should be evaluated and managed in the future instead of in the past. These companies would have opened up a lot of opportunities if they use AI to support talent management activities (Howard, 2019).

Resource performance is a vital aspect of an industrial process to enable performance verification and works as a crucial game changer in the changing asset management environment. Smart automation technologies offer a new approach to personnel management and boosting corporate performance, as well as other areas of performance management. AI is more than simply a trendy term in every business. This new technology is already delivering chances to infuse AI into performance management operations, from recruiting to employee experience and learning (Khang, Rani & Sivaraman, 2022).

Traditional assessment systems are inefficient, restrictive, and time-consuming. According to the AIOCF model (Khang, 2024), traditional performance management takes roughly 210 hours, or five weeks of managers' time to analyze performance. AI has been adept at processing data in a variety of industries, including healthcare (Khang et al., 2022), business, and transportation infrastructure systems, among other AI-enabled benefits. There's no knowing how AI will change the way businesses manage their employees' performance.

When used to measure key performance indicators and make more data-driven choices, AI can greatly enhance performance management. It may also be integrated into the performance management process to assist with analysis, provide metrics, and make recommendations. (Heena, 2022). Companies that have long exemplified the "stack and rank" strategy, such as General Electric (GE) and Microsoft (Stephen, 2013; Catherine, 2023), have done away with their yearly systems for grading and assessing personnel in favor of exploring new approaches that provide continuous feedback and coaching.

Netflix no longer measures its employees against annual goals, since such goals have become more flexible and subject to fast change. Google has changed the way it pays strong achievers at all levels. Many review processes that managers usually conduct manually have been automated by the use of AI (Rani et al., 2022).

AI-powered performance management is a business favorite. It allows them and their managers to focus on real data and execute employee performance appraisals. Employees need to be able to work together and make decisions. Executives can now better analyze the effects of such collaboration thanks to AI technology under the process optimization, as a result, it can lead to a better TM approach (Prentice, Weaven & Wong, 2020).

9.2.4 AI AND THE ONBOARDING PROCESS

The onboarding process for new employees is the first impression they get of your company, and it also can either strengthen or weaken that perception. The onboarding process begins as soon as you make an offer, and it continues until the employee is proficient at carrying out their job responsibilities.

Onboarding includes almost all of the actions that take place at that time, such as the orientation session, training schedule, installation of performance metrics, and feedback loop. The procedure is essential, since it reveals the company's work culture and symbolizes inculcation. The adoption of AI in the onboarding process has helped businesses accomplish the 4cs: compliance, clarity, culture, and connection.

Machine learning facilitates maintaining uniform standards for employee onboarding, reduces workload time, speeds up the onboarding process, and fosters a good onboarding experience. It also helps to maintain a smooth, streamlined procedure (Creegan & Roberts, 2021).

Onboarding using AI lessens discriminatory practices, which are caused by commonplace human preconceptions, preferences, beliefs, and worldviews that skew our judgment. There are four stages in which artificial intelligence has supported onboarding procedures, such as a coach, monitor, helper, or teammate. High-performing businesses have embraced the “employee first” philosophy and applied customer experience ideas to TM and development in HR (Cai et al., 2021).

To provide an improved employee experience, AI and human intelligence are frequently integrated. AI aids in the automation of several repetitive HR duties as well as the redesign of procedures. Due to rising digitization and remote work, the workforce has seen tremendous transformation in recent years.

As a result, employees now have different expectations and are seeking applications that are simple to use and well-designed, so they can feel more involved in their jobs and make their duties simpler (Dhanoa et al., 2022). AI takes care of some of the mundane manual duties that HR used to do, such as distributing paperwork, setting up onboarding sessions, and answering some simple inquiries from new hires throughout the onboarding process.

By using this technique, new hires receive prompt answers to their questions without experiencing any delays throughout the onboarding process. New hires’ onboarding experiences may be enhanced with onboarding chatbots by giving them improved communication flow and on-demand access to vital information. The effectiveness of your company’s onboarding procedures has a big impact on how well it does business. A solid onboarding procedure may increase recruit retention by much as 82%, as per a recent Glassdoor team’s (2009) survey. Additionally, it can boost productivity by much as 70%.

AI eliminates the need for onboarding by taking place during regular business hours or at a set office location. A recruit may be guided through every step of onboarding by AI and chatbots, which are also capable of responding to queries as they come up. AI has become a genuine partner to HR in managing talents in the workplace, enhancing human intuition and creativity while simultaneously teaching the machine how quickly to analyze a wide range of data in enormous numbers (Makarius et al., 2020).

Many HR professionals are investing in things like predictive analytics, artificial intelligence, and other automated processes, in an effort to sift through thousands of CVs from Internet apps to recruit new employees for their organization (Bhardwaj, Singh & Kumar, 2020).

An article published in *The Times of India* (TOI, 2023) stated that there has been a shift in the work environment due to the sudden presence of the pandemic in our daily lives. The challenge was to maintain and manage an efficient workforce.

AI helps in onboarding processes such as boarding fraud detection, because with virtual hiring and remote employment, employee moonlighting has grown prevalent. Integrating AI technologies like face recognition and likeness monitoring into

onboarding programs may help keep a check on this and prevent an organization from productivity loss, exploitation of corporate resources, conflicts of interest, potential data leakage, and documentation security since this data may be generated from a variety of sources.

However, to extract usable information from this data, it must be processed and formatted. By automatically evaluating files and linking them to important information, an AI-enabled system for document management assists enterprises in using this data.

AI-powered document management solutions improve document security by recognizing sensitive and personally identifiable information so that users may handle the material with caution, because most piratical verification (AI assists in real-time personnel verification through the use of technology like picture detection and biometric authentication).

AI-powered verification shortens onboarding times and enhances the onboarding experience by eliminating the need for multiple follow-ups, physical documents, and physical presence, leading to a smooth flow of managing talent (Tambe & Tambay, 2020; Trehan, 2022).

9.3 IMPACT OF AI ON GLOBAL ENTERPRISES: AI IN PRACTICE

All the digital giants in the world, from Alibaba to Amazon, are vying to be the world leaders in AI. These companies are AI trailblazers, applying AI to deliver contemporary goods and services. Here are some of the best examples of how these companies are using artificial intelligence.

9.3.1 ALIBABA

Alibaba is the world's largest online forum, with more sales than Amazon and eBay combined. AI is used in Alibaba's daily conveyance to predict what customers want to buy. The company uses natural language processing to unquestionably generate product data for its website.

Alibaba has also used AI to build smart cities with its City Brain initiative. This initiative uses AI algorithms to reduce traffic congestion by tracking every vehicle in the city.

In addition, through its cloud computing arm called Alibaba Cloud, Alibaba adopted AI to help agriculturalists monitor crops, increase production, and reduce costs (Rani et al., 2023).

9.3.2 ALPHABET INC. – GOOGLE

Self-driving technology company Waymo was founded as an initiative of Google. Waymo is now unable to charge fares and must rely on a human driver. Google is integrating machine learning and artificial intelligence into its operations. An AI voice system can make a phone call and arrange bookings using natural language.

9.3.3 AMAZON

Amazon is present in many facets of the company as well as its AI business with its digital language assistant Alexa bot, and another unique way Amazon uses AI is to

show you items before you even consider buying them. They say they can collect all the data about people's shopping behavior, use that data to recommend products to consumers, and even use predictive analytics to predict what they're buying needed when they need it.

We're sure at a time when many brick-and-mortar retailers are struggling to stay relevant, America's largest electronics retailer has come up with a new concept for convenience stores: Amazon Go. No need to verify as in other stores. AI devices in the store record what you pick up, then claim those items through the Amazon Go app on your phone. There's no cash register, so you bring your own carrying case to put things in, and the camera tracks every step to find every item you put in your pocket.

9.3.4 APPLE

Apple is a global high-tech company that sells consumer products such as iPhone smartwatches, as well as office software and internet services. Apple adopted AI and machine learning (ML) in devices such iPhones with Face ID, as well as products like AirPods, Apple Watch, and HomePod smart speakers enabled by Siri.

Apple is also extending its services by using AI to endorse music in Apple Music, find photos in iCloud, and maneuver to your next engagement with Maps.

9.3.5 BAIDU

Baidu has implemented AI in several ways and has software called Deep Voice that uses AI and deep learning to replicate voices in just 3.7 seconds of speech. They used the same technology to create an author's voice reading gadget with everything is automatic, and no recording studio is needed.

9.3.6 IBM

IBM has been a leader in AI for many years. It has been more than 20 years since IBM's Deep Blue machine beat the first human-world chess unconquered. The company has since used Watson computers to win other human-versus-machine games, including the game show *Jeopardy!*

Project Debater is IBM's latest achievement in AI. This is a percipient computing engine that has battled competent debaters and advanced human-like interaction.

9.4 CONCLUSION

AI has become a part of our daily lives, whether we're making routine purchases on Flipkart or Myntra, or watching Netflix or YouTube for entertainment. We have allowed AI to creep into our lives, either directly or indirectly (Khang et al., 2022).

Organizations have not been left out of the effect of AI. Artificial neural networks learn how to categorize sounds, pictures, text, or other input by recognizing features within a training dataset through a process called backward propagation. The area where AI ends up having the most immediate influence is TM (Vrushank, Khang & Rani, 2023).

The workplace and the worker are significantly impacted globally by the rapid changes in demography, technology, and globalization. The companies are focusing on TM and its effective functioning through the adaptation of AI. The demands of employees are being met by progressive businesses across all industries in their people-management strategies along with implementing a smart way of working with AI.

Whether we are apprehensive about it or not, artificial intelligence permeates our daily lives and is everywhere. AI works in the background every time you check your Facebook news feed, use Google, get Amazon product recommendations, or book a trip online. Before accepting a job offer, employees look at their potential for future growth. They seek companies that are expanding at the same speed as the market and have efficient TM systems (Morris et al., 2023).

One in four individuals would not accept a position where the company's ideals conflict with their own. As a result, if an employee needs to choose between several job offers, they are more inclined to join a firm using AI to show their long-term employability. With AI serving as a real buddy in several aspects of TM, the task goes well.

The best uses of AI in the human resources segment involve humans and technology collaborating to achieve organizational goals and attain better TM practices. This may be the most significant point to consider for an organization trying to employ AI (Khang et al., 2023).

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